SmolVLM for Dense Video Captioning

Week 7: Internship Update

- Implementation of (https://github.com/marija-brkic/Video-Model-Research.git):
    - MaxInfo algorithm
    - CSTA algorithm (Video Summarization task, CNN-based Spatiotemporal Attention)

Algorithm 1 MaxInfo Block: SVD + MaxVol for Keyframe Selection

Input: A set of n frames $\mathbf{I} = \{i_1, i_2, \ldots, i_n\}$

Embedding: Convert each frame i_j into a [CLS] embedding:

$$q_n = \text{flatten}(\text{clip\_model}(i_n)), \mathbf{Q} = \begin{bmatrix} q_1 \\ q_2 \\ \vdots \\ q_n \end{bmatrix} \in \mathbb{R}^{n \times d}.$$

SVD Reduction: Perform truncated SVD on $\mathbf{Q}$:

$$\mathbf{Q} \approx \mathbf{U}_r \Sigma_r \mathbf{V}_r^{\mathsf{T}} \quad \rightarrow \quad \mathbf{Q}_s = \mathbf{U}_r \in \mathbb{R}^{n \times r}.$$

MaxVol Selection: Run $\text{rect\_maxvol}(\mathbf{Q}_s, \text{tol})$ to find pivot indices:

$$\text{piv} = \text{rect\_maxvol}(\mathbf{Q}_s, \text{Tol}),$$

identifying rows (frames) that span the reduced embedding space.

Output: Indices **piv** of the most informative keyframes.

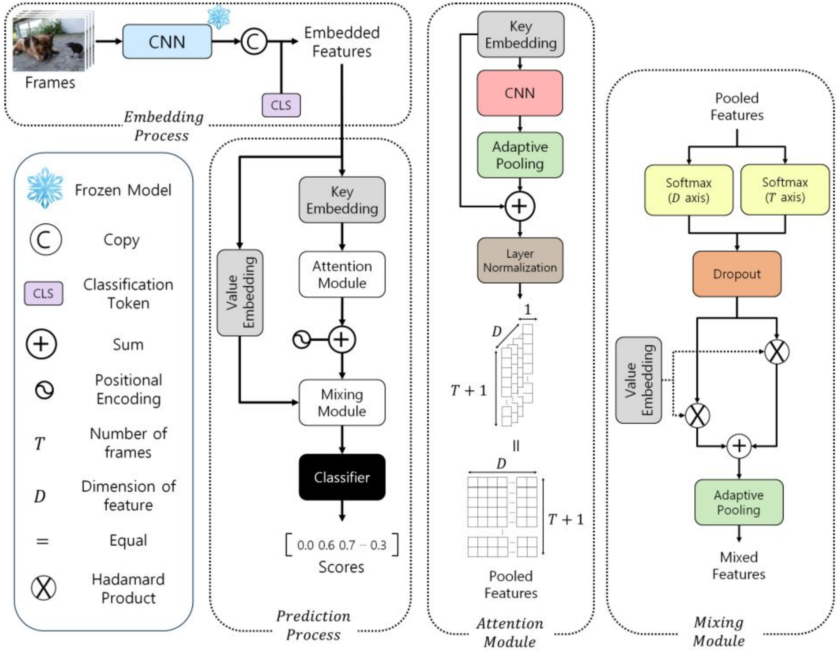


Figure 3. Architecture of CSTA

MaxInfo: A Training-Free Key-Frame Selection Method Using Maximum Volume for Enhanced Video Understanding

- Fast and Slow version
  - Fast version: MaxInfo is applied directly to the same number of frames as the original uniform sampling.
  - Slow version: A larger pool of frames ($N \gg n$) is initially sampled to ensure extensive coverage. The MaxInfo Block is then applied to select the most diverse frames.
- If the selected number of frames is higher than the allowed input of a model, frames are uniformly sampled

MaxInfo: A Training-Free Key-Frame Selection Method Using Maximum Volume for Enhanced Video Understanding

- Scene-aware MaxInfo
  - For each chunk, they extract CLIP embeddings, apply SVD for dimension reduction, and run MaxVol to select the most representative frames

$$\mathbf{I} = \bigcup_{i=1}^{32} \mathbf{I}^{(i)}, \quad \mathbf{I}^{(i)} = \{i_j | j \in \text{chunk } i\}.$$

  - PySceneDetect: ContentDetector, default parameters (threshold=27.0, min_scene_len=15)
  - CLIP embeddings in parallel
  - Should different embedding be tested?
  - More frames extracted with scene-aware algorithm

MaxInfo: A Training-Free Key-Frame Selection Method Using Maximum Volume for Enhanced Video Understanding

- Some additional results
  - Slow mostly better
  - For shorter videos uniform sampling might perform better

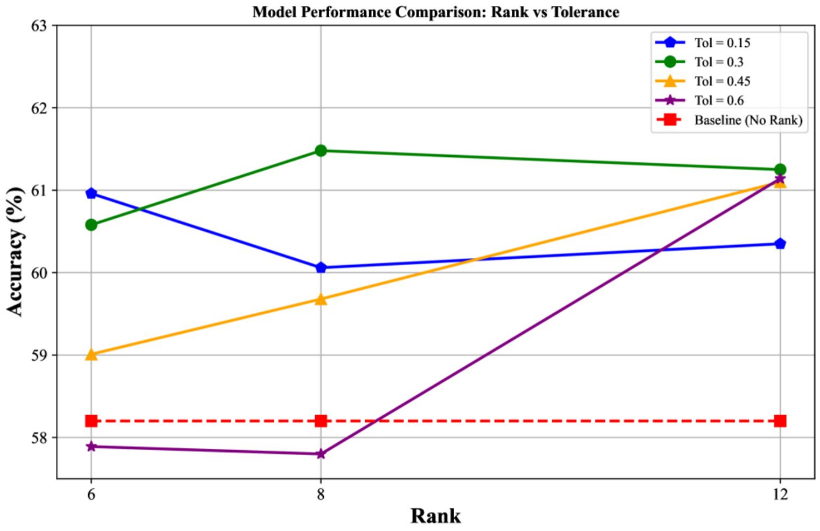
MODEL	INPUT MAX. FRAMES	OUTPUT FRAMES RANGE	AVG. FRAMES	ACC.
INTERNVL2 1B (Chen et al., 2024)	32	-	32	43.38
+ MaxInfo$_{fast}$	32	(1, 32)	30	43.6 +0.22%
+ MaxInfo$_{slow}$	64	(1, 16)	16	44.65 +1.05%
INTERNVL2 2B (Chen et al., 2024)	32	-	32	46.97
+ MaxInfo$_{fast}$	32	(1, 32)	30	47.27 +0.73%
+ MaxInfo$_{slow}$	64	(1, 16)	16	46.82 -0.15%
LLAVA-VIDEO 7B (Zhang et al., 2024c)	64	-	64	58.2*
+ MaxInfo$_{fast}$	64	(1, 64)	52	60.21 +2.01%
+ MaxInfo$_{slow}$	128	(1, 64)	58	61.48 +3.28%
LLAVA-VIDEO 72B (Zhang et al., 2024c)	64	-	64	61.9*
+ MaxInfo$_{fast}$	64	(1, 64)	52	65.37 +3.47%
+ MaxInfo$_{slow}$	128	(1, 64)	58	64.92 +3.02%

MODEL	SIZE	MVBENCH	EGOSCHEMA	LONGVIDEOBENCH
INTERNVL2 (Chen et al., 2024)	1B	55.42	34.02	43.38
+ MaxInfo$_{(128,32)}$	1B	**56.07**	34.11	**42.78**
△		+0.65%	+0.09%	-0.6%
INTERNVL2 (Chen et al., 2024)	2B	59.72	46.10	43.97
+ MaxInfo$_{(128,32)}$	2B	59.02	46.33	**47.27**
△		-0.7%	+0.23%	+3.3%

MODEL	SIZE	ACC.
QWEN2-VL (Wang et al., 2024a)	2B	52.18
+ MaxVol	2B	$\boxed{52.37}$
+ Chunk-Based	2B	**52.69**
QWEN2-VL (Wang et al., 2024a)	7B	64.32
+ MaxVol	7B	$\boxed{64.59}$
+ Chunk-Based	7B	**64.82**

MaxInfo: A Training-Free Key-Frame Selection Method Using Maximum Volume for Enhanced Video Understanding

- Hyperpameter testing: tolerance and rank
- LLaVA-Video-7B, 96 initial frames
- 3.3% improvement with tol = 0.3 and R = 8 (that is what I used)



MaxInfo: A Training-Free Key-Frame Selection Method Using Maximum Volume for Enhanced Video Understanding

- Tolerance and rank guidelines:

1. **Large Rank, Moderate Tol:** When the Rank R is large, a moderate Tolerance (Tol) helps avoid selecting overly similar frames, optimizing the trade-off between diversity and redundancy.

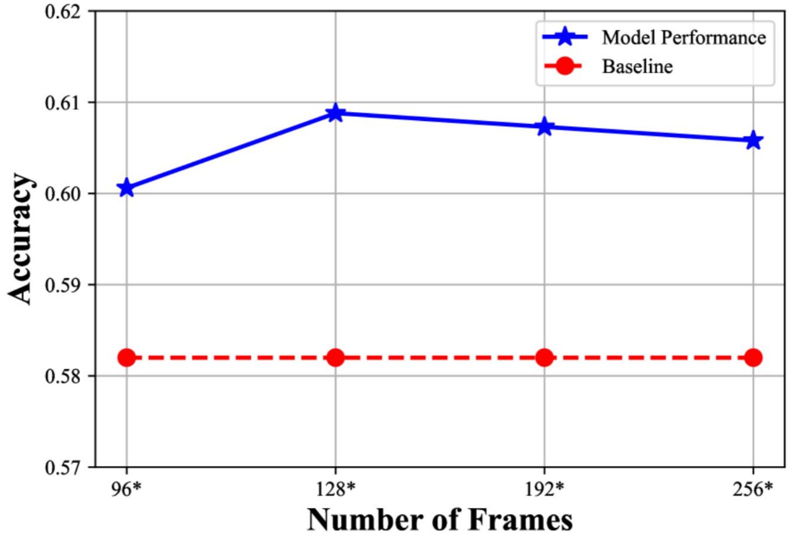
2. **Balanced Selection:** An excessively large R combined with a very small Tol may cause the frame set to grow excessively, potentially exceeding the model's capacity. Thus, R and Tol should be selected based on the maximum feasible input frame count for a given LLM.

3. **Performance Sensitivity:** Our experiments show that performance is most sensitive to Tol values in the range $\text{Tol} \in [0.1, 0.6]$. Beyond this range, improvements plateau or regress due to over-pruning or under-pruning.

4. **Convergence:** When setting $R = 12$ and varying Tol values to 0.30, 0.45, and 0.60, the best result converged at 61.1%.

MaxInfo: A Training-Free Key-Frame Selection Method Using Maximum Volume for Enhanced Video Understanding

- Comparison of initial number of frames:



CSTA: CNN-based Spatiotemporal Attention for Video Summarization

- Adjusted the original algorithm for frame sampling
- Adjustable for both the original algorithm (scoring of every 15th frame, video spliting) and for scoring every frame
- Not perfect for shorter videos
- $M \in R^{(T+1)xD}$
- After adaptive pooling: $M' \in R^{TxD}$
- $R \in R^{TxD}$
- $S \in R^{T}$

$$R = LayerNorm(Dropout(ReLU(FC(M'))))$$

$$S = Sigmoid(FC(R))$$

Algorithm 1: *Prediction Process*

input : $E \in \mathbb{R}^{3\times(T+1)\times D}$
output: $S \in \mathbb{R}^{T}$
begin
  $E^K = W^K E$
  $E^V = W^V E[0]$

  $P = Attention\ Module(E^K)$ Section 3.4
  $P_{pos} = P + Positional\ Encoding$
  $M = Mixing\ Module(P_{pos}, E^V)$ Section 3.5
  $S = Classifier(M)$
  return S
end